

Solution Requirements

Date	28 June 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely register and log in using their email and password to access the crop recommendation system.</i>	High
2	Authorization levels	<i>Users can submit soil data and view recommendations, while the administrator can manage datasets, machine learning models, and user records.</i>	High
3	External interfaces	<i>The system uses the crop dataset and machine learning libraries such as Scikit-learn.</i>	Medium

		<i>It can be extended to integrate weather APIs in future versions.</i>	
4	Transactions processing	<i>The system accepts soil parameters, processes them using the trained machine learning model, stores prediction results, and generates recommendations.</i>	High
5	Reporting	<i>Generates crop recommendation reports containing the predicted crop, confidence score, and farming recommendations.</i>	High
6	Business rules, etc.	<i>Predictions are generated only after all mandatory soil parameters are entered and validated. The recommendation is based on the trained machine learning model.</i>	High
7	Compliance to laws or regulations	<i>User information is stored securely and used only for agricultural prediction purposes while maintaining data privacy.</i>	Medium
8	Other	<i>Users can view previous prediction results and download recommendation reports for future reference.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should process user input and generate crop predictions quickly.</i>	<i>Prediction generated within 2 seconds.</i>
2	Scalability	<i>The application should support increasing numbers of users and larger agricultural datasets without affecting performance.</i>	<i>Supports 1,000+ users and scalable dataset storage.</i>
3	Security & Data Privacy	<i>User information and prediction data should be stored securely with proper authentication and access control.</i>	<i>Secure login, password protection, and restricted access.</i>
4	Reliability & Availability	<i>The application should be available whenever users need to perform crop prediction with minimal downtime.</i>	<i>99% system availability during normal operation.</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy for farmers and agricultural users to operate</i>	<i>User-friendly interface with clear navigation and responsive design.</i>
6	Other	<i>The system should be maintainable, portable, and easy to update with new datasets or machine learning models.</i>	Modular code structure and compatibility with Windows/Linux environment